Vibration Control of an Overhead Crane with Hoisting Motion using Input Shaping Technique

Ho Duc Tho[1], Kazuhiko Terashima[2], and Takanori Miyoshi[1]

***Abstract*— In this paper, an input shaping-based control law is designed for an overhead crane system with time-varying cable length. The proposed method employs a zero vibration (ZV) input shaper. It is well-known that the ZV input shaper is not robust to modeling errors causing by the time-varying dynamics, which leads to the fact that a direct application of ZV input shaper on time-varying systems often results in poor vibration suppression performance. Therefore, we first apply the standard ZV input shaper on an undamped linear time-invariant (LTI) second-order system. This step provides us a reference oscillation function that is exactly zero after the trolley stops. The real control input is then devised to perfectly match the real vibration of the crane system with the reference oscillation, and hence a vibration-free transportation can be achieved. The unshaped command is parameterized and its coefficients are designed to deal with control objectives other than the vibration elimination. The simulation results are provided to validate the efficacy of the proposed approach.**

I. Introduction

Cranes are the machines that can lift, lower, and move the payload from one point to another within a workspace. They are typically used for transferring enormous payloads weighing hundreds of kilograms to dozens of tons, which are well beyond a human's natural capabilities. There are several types of cranes, in which overhead cranes are widely accepted as the most important means of transportation in factories, in storage yards, at harbors, and so on. In an overhead crane, the load is hung below a trolley by ropes and its height can be adjusted by hoisting up or down. The trolley is actuated by an electric motor to relocate the payload in a linear fashion. Since the rope is flexible, the suspended payload is highly prone to vibrate during and after a maneuver. This unwanted vibration drastically lessen the safety and productivity of the transportation. Therefore, three control objectives are required: 1) bring the trolley to a desired destination, 2) lift up or lower the load down to a prescribed height, and 3) cease the swing movement of the payload. This is a demanding task because the overhead crane is an underactuated system, with only two control inputs (trolley and hoisting actuation) and three independent outputs to regulate (the swing angle is the unactuated degree of freedom). Consequently, over the past decades, considerable interests were mounting on the vibration control for the overhead crane system with hoisting motion. They can be divided into two main categories: feedback and feedforward controls. Feedback control techniques have demonstrated their outstanding properties in coping with disturbances and uncertainties. Nonetheless, they necessitate a complex and costly sensor structure that requires regular maintenance. This is one of the reasons why feedback controllers are still rarely used in practical cranes. In numerous contexts, feedforward controls are more desirable for industry applications due to their simplicity, ease of implementation, and no additional measurement system being required.

Among the feedforward control methods [1]–[5], the input shaping [6]–[8] is the most recognized one. Its fundamental idea is to convolute the original (or *unshaped*) command with a sequence of impulses called *input shaper*. The input shaper is designed in such a way that the vibration induced by all impulses is canceled out by themselves. The product of this convolution process is named *shaped command*, and it is the actual reference applied to the system. The input shaping method is particularly effective for vibratory LTI systems. Nevertheless, the application of input shaping controller on time-varying systems is not very successful in terms of vibration suppression. The reason is that the input shaper is specifically intended for a second-order LTI system. Any attempts to apply the input shaping concept on other systems without further investigations often lead to poor performance in vibration elimination. In cranes, the hoisting motion introduces a time-varying dynamics into the system. Singhose et al. [9] reported deleterious effects of the hoisting motion to performance of the input shaping controller on a gantry crane. The results claimed that an exact zero vibration cannot be guaranteed after the transportation. Until now, when using the input shaping control, the main method to deal with the varying cable length is to use robust input shaper types such as zero derivatives reinforcement [6], [10], extra insensitivity (EI) [7], [10]–[12], and optimized specified insensitivity (SI) [13], [14]. Recently, Ramli et al. [15] attempted to use an artificial neural network (ANN) trained by a particle swarm optimizing mechanism to modify the unity-magnitude ZV input shaper online. The major purpose is to handle the unpredictability of the rope length as well as the payload mass uncertainty. Despite this, none of the existing studies using input shaping have been able to totally eliminate vibration for overhead crane systems with time-varying cable length. This is the key issue that will be addressed in this paper.

A new input shaping-based controller for an overhead crane with hoisting motion will be developed to tackle the aforementioned challenge. By using the newly developed

[1]Ho Duc Tho and Takanori Miyoshi are with Department of System Safety, Nagaoka University of Technology, 1603-1 Kamitomiokamachi, Niigata, Japan. E-mails: hoductho@vos.nagaokaut.ac.jp, miyoshi@vos.nagaokaut.ac.jp

[2]Kazuhiko Terashima is with Toyohashi University of Technology, 1-1 Hibarigaoka, Aichi, Japan. E-mail: terasima@tut.ac.jp

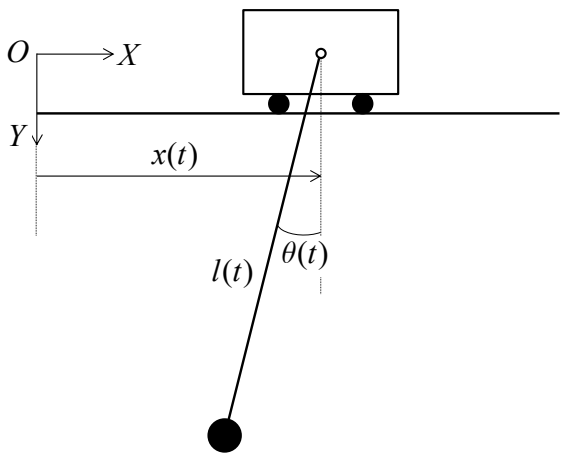

Fig. 1. Mathematical model of an overhead crane with hoisting motion.

approach, we can completely suppress vibration for a time-varying crane system using a non-robust ZV input shaper. As a result, the transfer time is significantly reduced as compared to conventional robust input shapers, which are slow and cannot guarantee a perfect ZV condition.

II. Problem formulation

The mathematical model of an overhead crane system is shown in Fig. 1. The trolley position, the swing angle, and the time-varying rope length are denoted as $x(t)$ (m), $\theta(t)$ (rad), and $l(t)$ (m), respectively. Moreover, $g = 9.81$ (m/s^2) is the gravitational acceleration.

The dynamics (in the kinematic level) of the overhead crane with a hoisting motion can be established using the Euler-Lagrange method, and it is governed by

$$l\ddot{\theta} + 2\dot{l}\dot{\theta} + g\sin\theta = -\cos\theta\ddot{x}. \tag{1}$$

Typically, an input shaping controller requires a linear model, so we need to linearize the system (1) around its bottom equilibrium. Applying the small angle approximation on (1) yields the following result

$$l\ddot{\theta} + 2\dot{l}\dot{\theta} + g\theta = u, \tag{2}$$

where it is denoted that $u = -\ddot{x}$, i.e., the acceleration $-\ddot{x}$ is the control input of the system. The model (2) will be employed in the subsequent control design process. Throughout this paper, it is assumed that the payload starts from rest, namely that $x(0) = \dot{x}(0) = \dot{l}(0) = \theta(0) = \dot{\theta}(0) = 0$ [2], [3], [5]. Moreover, let $l(0) = l_0$ where $l_0 > 0$ is a known initial rope length.

The main aim is to design an input shaping-based control law $u(t)$ that brings the trolley position $x(t)$ and the rope length $l(t)$ to prescribed references, at which their velocities are zero and the vibration ceases. Mathematically, for some finite terminal time $\tau > 0$, these requirements are equivalent to the following conditions

$$\begin{aligned} &x(\tau) = x_d,\ \dot{x}(\tau) = 0, \\ &l(\tau) = l_d,\ \dot{l}(\tau) = 0, \\ &\theta(\tau) = 0,\ \dot{\theta}(\tau) = 0, \end{aligned} \tag{3}$$

where x_d and $l_d > 0$ are the constant reference values for the trolley and the rope length respectively.

III. Input shaping-based control law design

As explained in Section I, the input shaper should not be applied directly to the time-varying system (2) because this will result in poor vibration suppression performance. Alternatively, the standard input shaping controller is firstly applied to a reference model, which is a second-order LTI system. This step guarantees that the oscillation of the reference model (referred to as the reference oscillation) is exactly zero at the completion of the maneuver. The actual control input for the linear time-varying system (2) is then devised to perfectly match the vibration of the real system with that of the reference oscillation. The subsequent presentation will elaborate such a design concept.

Denote $\bar{l} > 0$ as a constant nominal rope length. An undamped reference model is given as follows

$$\bar{l}\ddot{\xi} + g\xi = \bar{u}(t). \tag{4}$$

The natural angular frequency of the reference oscillation $\xi(t)$ is $\omega = \sqrt{g/\bar{l}}$ and its natural period is $T = 2\pi/\omega$. The control input $\bar{u}(t)$ in (4) is a standard input shaping controller, that is to say

$$\bar{u}(t) = u_r(t) * I(t), \tag{5}$$

where $*$ stands for the convolution operator. Moreover, $I(t)$ is the ZV input shaper with $\delta(.)$ is the Dirac delta generalized function [6]

$$I(t) = 0.5\left[\delta(t) + \delta(t - T/2)\right]. \tag{6}$$

To cope with control objectives other than the vibration suppression as shown in (3), the unshaped command $u_r(t)$ in (5) is parameterized as follows

$$u_r(t) = \begin{cases} \beta_1 t + \beta_0, & t \in [0,\ t_f] \\ 0, & t > t_f, \end{cases} \tag{7}$$

where β_i $(i = 0, 1)$ are to-be-determined variables and t_f is a positive number chosen by the designer. When the input (5) is applied on the reference model (4), the reference oscillation $\xi(t)$ is suppressed, namely that

$$\xi\left(t_f + T/2\right) = \dot{\xi}\left(t_f + T/2\right) = 0.$$

The next step is to design the real control input $u(t)$ in (2) to match the actual vibration $\theta(t)$ with the reference oscillation $\xi(t)$, i.e., we wish to make

$$\theta(t) \equiv \xi(t),\ \dot{\theta}(t) \equiv \dot{\xi}(t),\ \ddot{\theta}(t) \equiv \ddot{\xi}(t),\ \forall t \geq 0. \tag{8}$$

By using (2) and (4), we have

$$\begin{aligned} \ddot{\xi} &= -\frac{g}{\bar{l}}\xi + \frac{\bar{u}(t)}{\bar{l}} \\ \ddot{\theta} &= -2\frac{\dot{l}}{l}\dot{\theta} - \frac{g}{l}\theta + \frac{u(t)}{l}. \end{aligned} \tag{9}$$

Applying (9) on (8) yields

$$-2\frac{\dot{l}}{l}\dot{\xi} - \frac{g}{l}\xi + \frac{u(t)}{l} = -\frac{g}{\bar{l}}\xi + \frac{\bar{u}(t)}{\bar{l}},$$

from which the real control input $u(t)$ is determined by

$$u(t) = g\left(1 - \frac{l}{\bar{l}}\right)\xi + 2\dot{l}\dot{\xi} + \frac{l}{\bar{l}}\bar{u}. \tag{10}$$

The control law (10) when applying on the time-varying system (2) will ensure an exact zero residual vibration at the terminal time $t_f + T/2$, that is to say

$$\theta\left(t_f + T/2\right) = \dot{\theta}\left(t_f + T/2\right) = 0. \tag{11}$$

Therefore, the requirement of vibration suppression is now fulfilled. To further realize conditions regarding to the hoisting motion, the rope length function $l(t)$ is chosen as a third-order time polynomial as follows

$$l(t) = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3. \tag{12}$$

The coefficients $\alpha_i \in \mathbb{R}$ $(i = 0, \ldots, 3)$ in (12) are given by

$$\alpha_0 = l_0,\ \alpha_1 = 0,\ \alpha_2 = \frac{3\left(l_d - l_0\right)}{\left(t_f + T/2\right)^2},\ \alpha_3 = \frac{-2\left(l_d - l_0\right)}{\left(t_f + T/2\right)^3}, \tag{13}$$

to guarantee the essential initial conditions $l(0) = l_0$, $\dot{l}(0) = 0$ and the terminal conditions $l\left(t_f + T/2\right) = l_d$, $\dot{l}\left(t_f + T/2\right) = 0$.

Since the control input $u(t)$ is the acceleration of the trolley, we have to perform integration on (2) to determine constraints for which the remaining transport requirements $\dot{x}\left(t_f + T/2\right) = 0$ and $x\left(t_f + T/2\right) = x_d$ are fulfilled. To do so, an explicit form of the reference oscillation $\xi(t)$ needs to be known in advance. For this subtask, Laplace transform can be implemented. Denote s as the complex variable and $\mathcal{L}\{(.)(t)\}$ as the Laplace operator applied on the function $(.)(t)$. The Laplace transform of the unshaped command $u_r(t)$ is given by

$$U_r(s) = \mathcal{L}\left\{u_r(t)\right\} = \int_0^{t_f} \left(\beta_0 + \beta_1 t\right) e^{-st} dt, \tag{14}$$

where (7) has been used. Conducting integration by part on (14) yields

$$U_r(s) = \frac{\beta_0}{s} + \frac{\beta_1}{s^2} - \left(\frac{\beta_0 + \beta_1 t_f}{s} + \frac{\beta_1}{s^2}\right) e^{-st_f}. \tag{15}$$

Moreover, in the s-domain, the ZV input shaper (6) has a form of

$$\mathcal{I}(s) = 0.5\left(1 + e^{-sT/2}\right).$$

Performing Laplace transform on both sides of (5) results in $\bar{U}(s) = U_r(s)\mathcal{I}(s)$, where $\bar{U}(s)$ is the Laplace transform of $\bar{u}(t)$. By applying the operator $\mathcal{L}$ on (4), we can compute $\Xi(s)$, which is the Laplace transform of the reference oscillation $\xi(t)$, as follows

$$\begin{aligned} &\Xi(s)\left(\bar{l}s^2 + g\right) = \bar{U}(s) \\ \Longrightarrow\ &\Xi(s) = \frac{1}{2}\left[F_1(s) - F_2(s)e^{-st_f}\right]\left(1 + e^{-sT/2}\right). \end{aligned} \tag{16}$$

In (16), the terms $F_1(s)$ and $F_2(s)$ are given by

$$\begin{aligned} F_1(s) &= \frac{\beta_0/s + \beta_1/s^2}{\bar{l}s^2 + g} \\ F_2(s) &= \frac{\left(\beta_0 + \beta_1 t_f\right)/s + \beta_1/s^2}{\bar{l}s^2 + g}. \end{aligned} \tag{17}$$

To determine $\xi(t)$ from $\Xi(s)$, the following inverse Laplace transforms of $F_1(s)$ and $F_2(s)$ are needed

$$f_1(t) = \frac{\beta_0 + \beta_1 t}{g} - \frac{\beta_0}{g}\cos\omega t - \frac{\beta_1}{g\omega}\sin\omega t$$

$$f_2(t) = \frac{\beta_0 + \beta_1\left(t + t_f\right)}{g} - \frac{\beta_0 + \beta_1 t_f}{g}\cos\omega t - \frac{\beta_1}{g\omega}\sin\omega t.$$

Taking the inverse Laplace transform for (16) and using the time shifting property results in

$$\begin{aligned} \xi(t) = \frac{1}{2}\big[&f_1(t) + f_1\left(t - T/2\right)H\left(t - T/2\right) \\ &- f_2\left(t - t_f - T/2\right)H(t - t_f - T/2) \\ &- f_2\left(t - t_f\right)H\left(t - t_f\right)\big], \end{aligned} \tag{18}$$

where $H(.)$ is the Heaviside function. Having the explicit formula for the reference oscillation $\xi(t)$, we now can integrate (2) to determine the trolley position $x(t)$ and the trolley speed $\dot{x}(t)$.

Since a perfect matching between the reference oscillation $\xi(t)$ and the real vibration of the system $\theta(t)$ are ensured by the control law (10), from (2), we have

$$\ddot{x} = -l\ddot{\xi} - 2\dot{l}\dot{\xi} - g\xi. \tag{19}$$

Integrating both sides of the above equation and using the initial condition $\dot{x}(0) = 0$, the trolley speed $\dot{x}(t)$ is given by

$$\dot{x}(t) = -\int_0^t l(\sigma)\ddot{\xi}(\sigma)d\sigma - 2\int_0^t \dot{l}(\sigma)\dot{\xi}(\sigma)d\sigma - g\int_0^t \xi(\sigma)d\sigma. \tag{20}$$

Using the initial condition $\dot{\xi}(0) = \dot{\theta}(0) = 0$, the first term of the right-hand side (RHS) of (20) can be further evaluated by conducting integration by part as follows

$$\int_0^t l(\sigma)\ddot{\xi}(\sigma)d\sigma = l(t)\dot{\xi}(t) - \int_0^t \dot{l}(\sigma)\dot{\xi}(\sigma)d\sigma. \tag{21}$$

Substituting (21) into (20) results in

$$\dot{x}(t) = -l(t)\dot{\xi}(t) - \int_0^t \dot{l}(\sigma)\dot{\xi}(\sigma)d\sigma - g\int_0^t \xi(\sigma)d\sigma. \tag{22}$$

Continuing to perform integration by part on the second term of the RHS of (22) and using the initial condition $\xi(0) = \theta(0) = 0$, the following result can be obtained

$$\dot{x}(t) = -\dot{\kappa}(t) + \int_0^t \ddot{l}(\sigma)\xi(\sigma)d\sigma - g\int_0^t \xi(\sigma)d\sigma, \tag{23}$$

where $\kappa(t) = l(t)\xi(t)$. From (12), we have $\ddot{l}(t) = 2\alpha_2 + 6\alpha_3 t$, so (23) becomes

$$\dot{x}(t) = -\dot{\kappa}(t) + \left(2\alpha_2 - g\right)\int_0^t \xi(\sigma)d\sigma + 6\alpha_3\int_0^t \sigma\xi(\sigma)d\sigma. \tag{24}$$

The trolley speed $\dot{x}(t)$ is now governed by (24). To further obtain the trolley position $x(t)$, we can integrate (24) and perform an analogous procedure as above, which eventually results in

$$x(t) = -l(t)\xi(t) + (2\alpha_2 - g)\, t \int_0^t \xi(\sigma)d\sigma + (-2\alpha_2 + 6\alpha_3 t + g) \int_0^t \sigma\xi(\sigma)d\sigma - 6\alpha_3 \int_0^t \sigma^2\xi(\sigma)d\sigma. \tag{25}$$

Using the results in (24)–(25) on the transport requirements $\dot{x}\left(t_f + T/2\right) = 0$ and $x\left(t_f + T/2\right) = x_d$ leads to

$$\begin{aligned} (2\alpha_2 - g) \int_0^{t_f+\frac{T}{2}} \xi(t)dt + 6\alpha_3 \int_0^{t_f+\frac{T}{2}} t\xi(t)dt = 0 \\ (2\alpha_2 - g) \int_0^{t_f+\frac{T}{2}} t\xi(t)dt + 6\alpha_3 \int_0^{t_f+\frac{T}{2}} t^2\xi(t)dt = -x_d. \end{aligned} \tag{26}$$

Having the closed-form formula for the reference oscillation in (18), it is straightforward to obtain the following results

$$\begin{aligned} \int_0^{t_f+\frac{T}{2}} \xi(t)dt &= \Phi_0\beta_0 + \Gamma_0\beta_1 \\ \int_0^{t_f+\frac{T}{2}} t\xi(t)dt &= \Phi_1\beta_0 + \Gamma_1\beta_1 \\ \int_0^{t_f+\frac{T}{2}} t^2\xi(t)dt &= \Phi_2\beta_0 + \Gamma_2\beta_1, \end{aligned} \tag{27}$$

in which Φ_i and Γ_i $(i = 0, 1, 2)$ are given by

$$\begin{aligned} \Phi_0 &= \frac{t_f}{g},\ \Gamma_0 = \frac{t_f^2}{2g}, \\ \Phi_1 &= \frac{t_f\left(\pi + \omega t_f\right)}{2g\omega},\ \Gamma_1 = \frac{t_f^2\left(3\pi + 4\omega t_f\right)}{12g\omega}, \\ \Phi_2 &= \frac{t_f\left(4\omega^2 t_f^2 + 6\pi\omega t_f + 6\pi^2 - 24\right)}{12g\omega^2} \\ \Gamma_2 &= \frac{t_f\left(3\omega^2 t_f^3 + 4\pi\omega t_f^2 + 3\pi^2 t_f - 12t_f\right)}{12g\omega^2}. \end{aligned}$$

Substituting (27) into (26), the below result can be deduced

$$\boldsymbol{A}\begin{bmatrix}\beta_0\\ \beta_1\end{bmatrix} = \begin{bmatrix}0\\ -x_d\end{bmatrix},$$

where the matrix $\boldsymbol{A}$ is given by

$$\boldsymbol{A} = \begin{bmatrix}(2\alpha_2 - g)\,\Phi_0 + 6\alpha_3\Phi_1 & (2\alpha_2 - g)\,\Gamma_0 + 6\alpha_3\Gamma_1 \\ (2\alpha_2 - g)\,\Phi_1 + 6\alpha_3\Phi_2 & (2\alpha_2 - g)\,\Gamma_1 + 6\alpha_3\Gamma_2\end{bmatrix}.$$

Therefore, two unknowns β_0 and β_1 of the unshaped command (7) can be computed by

$$\begin{bmatrix}\beta_0\\ \beta_1\end{bmatrix} = \boldsymbol{A}^{-1}\begin{bmatrix}0\\ -x_d\end{bmatrix}. \tag{28}$$

By this, the control design procedure is accomplished.

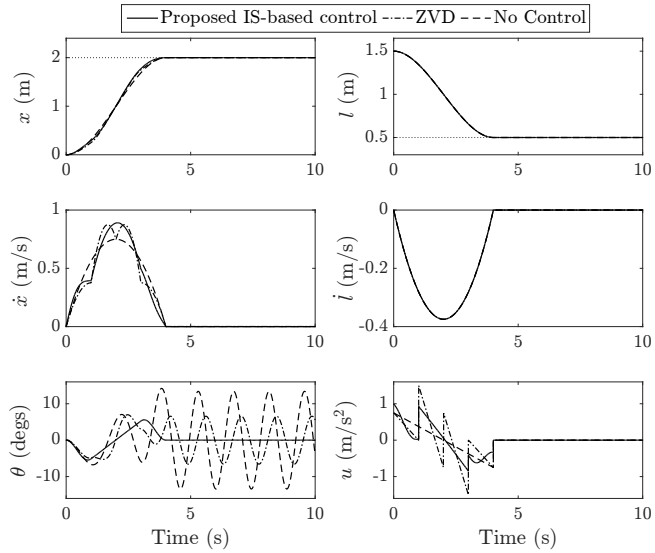

Fig. 2. The comparative simulation result in the hosting-up case.

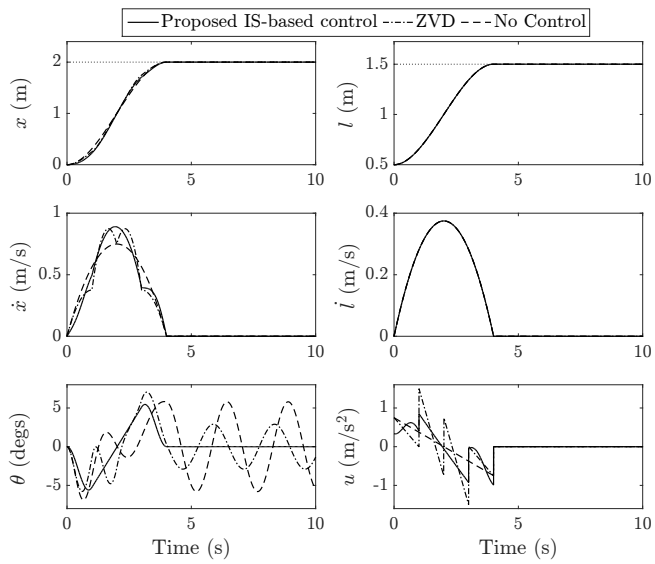

Fig. 3. The comparative simulation result in the hosting-down case.

In summary, the proposed input shaping-based control law is governed by (10), in which the rope length function $l(t)$ is given by (12) whose coefficients are shown in (13). Morever, the reference oscillation $\xi(t)$ can be obtained by (4). The term $\bar{u}(t)$ can be computed by (5), and the coefficients of the unshaped command $u_r(t)$ are given in (28).

IV. Simulation results

In this section, the proposed input shaping-based controller is compared with a no-control case and the standard robust input shaper ZVD (zero-vibration-derivative) [6]. The following two cases are considered

- Hoisting-up case: $l_0 = 1.5$ m, $l_d = 0.5$ m.
- Hoisting-down case: $l_0 = 0.5$ m, $l_d = 1.5$ m.

In both cases, the constant reference value for the trolley is $x_d = 2$ m. The proposed controller (10) employs a nominal

TABLE I

MAGNITUDES OF RESIDUAL VIBRATIONS IN THREE SCHEMES.

Case	Proposed control law (10)	ZVD	No-control
Hoisting-up	$0.012°$	$6.571°$	$13.403°$
Hoisting-down	$0.007°$	$2.898°$	$5.798°$

rope length of $\bar{l} = 0.5\,(l_0 + l_d)$. The terminal time t_f of the unshaped command $u_r(t)$ (see (7)) is chosen as $t_f = 3$ s. According to (13) and (28), essential coefficients for the rope length function $l(t)$ and the unshaped command $u_r(t)$ are

- Hoisting-up case: $\alpha_0 = 1.5$, $\alpha_1 = 0$, $\alpha_2 = -0.1872$, $\alpha_3 = 0.0312$, $\beta_0 = 1.3210$, $\beta_1 = -0.8891$.
- Hoisting-down case: $\alpha_0 = 0.5$, $\alpha_1 = 0$, $\alpha_2 = 0.1872$, $\alpha_3 = -0.0312$, $\beta_0 = 1.3464$, $\beta_1 = -0.8891$.

The computational time of both cases are 0.005 s on a personal computer with 2.7-GHz Intel Core i5-4310M and 8 GB of RAM using MATLAB. Therefore, it is possible to implement the proposed input shaping-based controller in a real time fashion.

In the no-control context, both trolley and hoisting motion employ the unshaped command $u_r(t)$ profiles shown in (7) to reach their reference values without considering the vibration suppression. Moreover, for the standard input shaping control scheme, a mildly robust input shaper ZVD is utilized, whose expression is given as follows

$$I_{\text{ZVD}}(t) = \frac{1}{4}\delta(t) + \frac{1}{2}\delta\left(t - \frac{T}{2}\right) + \frac{1}{4}\delta\left(t - T\right).$$

The unshaped command convolving with the ZVD input shaper has a similar form of $u_r(t)$ as shown in (7). The convolution product of the ZVD input shaper and $u_r(t)$ is the acceleration command applying directly to the time-varying crane system (2). The same rope length function $l(t)$ are used for all scenarios.

It is worth emphasizing that the transfer times of the three methods (i.e., the proposed control law (10), the no-control case, and the standard input shaping scheme using ZVD input shaper) are forced to be the same in order for the comparisons to be fair. As a result, the performance of each approach may be assessed using the amount of residual vibration at the end of transportation without bias.

The comparative simulation results are depicted in Fig. 2 and Fig. 3 for the hoisting-up and the hoisting-down case, respectively. It is remarked that the full time-varying nonlinear system (1) is employed in all simulations. It can be seen from Figs. 2–3 that the proposed controller is able to completely eliminate the residual oscillation after the maneuver. On the other hand, substantial vibrations persist in the case of standard input shaping method or no-control. To gain a better understanding of each control scheme's performance, the magnitudes of residual vibrations in all cases are reported in Table I. Note also that the maximum magnitude of the proposed control law is considerably smaller than the ZVD input shaping controller. Hence, under a relatively similar transportation time, the newly established approach outperforms the standard input shaping method.

V. CONCLUSION AND FUTURE WORK

This note introduced a new input shaping-based vibration controller for an overhead crane with time-varying rope length. Comparative simulation results suggested that the proposed approach is superior to the standard robust input shaping control using ZVD input shaper. Currently, the rope length was only assumed to be a third-order time polynomial and a non-robust ZV input shaper was utilized. In our future work, these factors will be generalized in order to achieve a more resilient and robust transportation. Moreover, experimental studies of the proposed approach on an actual overhead crane system are being prepared.

REFERENCES

[1] Y. Sakawa and Y. Shindo, "Optimal control of container cranes," *Automatica*, vol. 18, no. 3, pp. 257–266, 1982.

[2] K. Terashima, Y. Shen, and K. Yano, "Modeling and optimal control of a rotary crane using the straight transfer transformation method," *Control Engineering Practice*, vol. 15, no. 9, pp. 1179–1192, 2007.

[3] N. Uchiyama, H. Ouyang, and S. Sano, "Simple rotary crane dynamics modeling and open-loop control for residual load sway suppression by only horizontal boom motion," *Mechatronics*, vol. 23, no. 8, pp. 1223 – 1236, 2013.

[4] B. Kolar, H. Rams, and K. Schlacher, "Time-optimal flatness based control of a gantry crane," *Control Engineering Practice*, vol. 60, pp. 18 – 27, 2017.

[5] H. D. Tho, A. Kaneshige, and K. Terashima, "Minimum-time s-curve commands for vibration-free transportation of an overhead crane with actuator limits," *Control Engineering Practice*, vol. 98, p. 104390, 2020.

[6] N. C. Singer and W. P. Seering, "Preshaping command inputs to reduce system vibration," *ASME J. of Dynamic Systems, Measurement, and Control*, vol. 112, no. 1, pp. 76–82, Mar 1990.

[7] W. Singhose, "Command shaping for flexible systems: A review of the first 50 years," *Int. J. of Precision Engineering and Manufacturing*, vol. 10, no. 4, pp. 153–168, Oct 2009.

[8] H. D. Tho, N. Uchiyama, and K. Terashima, "Resonance-based tossing control for bulk materials transportation of an overhead crane," *IEEE Trans. on Industrial Electronics*, vol. 68, no. 1, pp. 609–621, 2021.

[9] W. Singhose, L. Porter, M. Kenison, and E. Kriikku, "Effects of hoisting on the input shaping control of gantry cranes," *Control Engineering Practice*, vol. 8, no. 10, pp. 1159 – 1165, 2000.

[10] J. Vaughan, A. Yano, and W. Singhose, "Robust negative input shapers for vibration suppression," *ASME J. of Dynamic Systems, Measurement, and Control*, vol. 131, no. 3, Mar 2009.

[11] W. Singhose, L. Porter, T. Tuttle, and N. Singer, "Vibration reduction using multi-hump input shapers," *ASME J. of Dynamic Systems, Measurement, and Control*, vol. 119, no. 2, pp. 320–326, Jun 1997.

[12] K. H. Rew, C. W. Ha, and K. S. Kim, "An impulse-time perturbation approach for enhancing the robustness of extra-insensitive input shapers," *Automatica*, vol. 49, no. 11, pp. 3425–3431, 2013.

[13] N. Singer and W. Seering, "An extension of command shaping methods for controlling residual vibration using frequency sampling," in *Proc. of IEEE Int. Conf. on Robotics and Automation*, vol. 1, 1992, pp. 800–805.

[14] L. Y. Pao and M. A. Lau, "Robust input shaper control design for parameter variations in flexible structures," *ASME J. of Dynamic Systems, Measurement, and Control*, vol. 122, no. 1, pp. 63–70, Oct 1999.

[15] L. Ramli, Z. Mohamed, and H. I. Jaafar, "A neural network-based input shaping for swing suppression of an overhead crane under payload hoisting and mass variations," *Mechanical Systems and Signal Processing*, vol. 107, pp. 484 – 501, 2018.